

MA 531: Introduction to Cryptography

Course Code: MA 531

Course Name: Introduction to Cryptography

L-T-P-C: 2-1-2-4

Prerequisites: IC 112 (for UG)

Intended for: UG/PG/PhD

Mutual Exclusion: None

Distribution: Elective

Approval: 65th BoA

Preamble

This course is intended to provide a foundation on the theory and practice of cryptography. Our goal is to learn about basic cryptographic techniques like encryption and message authentication in the private-key (symmetric) and public-key (asymmetric) settings, with the mathematical definition of security. We will also focus on learning thoroughly the mathematical concepts required to develop the foundation of secure cryptographic primitives. The topics in this course will include the following:

Course Modules

Lectures: 28 Hours

- **Unit 1: Introduction, Classical Cryptography, and Perfectly Secret Encryption:** Overview; Some simple cryptosystems (Shift cipher, Substitution cipher, Vigenère cipher, Permutation cipher etc.) and their cryptanalysis; one time pad, Shannon's theory of perfect security, limitations and other security notions. (5)
- **Unit 2: Private Key Encryption:** Modular arithmetic, Groups, Finite fields, pseudorandom functions, pseudorandom generators, stream ciphers, one-way functions, block ciphers and modes of operation, message authentication codes, some practical constructions. (5 lectures)
- **Unit 3: Hash Functions and applications:** Definition, collision resistance, Merkle–Damgård transform, Davis–Meyer construction, the sponge construction, SHA and HMAC. (4 lectures)
- **Unit 4: Public Key Encryption:** Some background mathematical concepts: Number fields, Primality testing and factoring, Discrete logarithms, Elliptic curves; Some computationally hard mathematical problems and their state-of-the-art; El Gamal, RSA and Rabin constructions, some attacks on RSA, Digital Signatures and Identification Schemes. (8 lectures)
- **Unit 5: Modern Aspects of Cryptography and Post-quantum Cryptography (basic notions):** Secret sharing; zero-knowledge proofs; multi-party computation; basics of post-quantum cryptography, with emphasis on lattice-based and code-based cryptography. (6 lectures)

Tutorials for discussing assignments: 14 Hours

Practical sessions for implementation using Python: 28 Hours



Text Books

(Latest, only 2)

1. Jonathan Katz and Yehuda Lindell, *Introduction to Modern Cryptography*, Chapman & Hall CRC Press, USA, 2007.
2. Avijit Das and C. E. Veni Madhavan, *Public-Key Cryptography: Theory and Practice*, Pearson Education, India, 2009.

References

1. Dan Boneh and Victor Shoup, *A Graduate Course in Applied Cryptography*, 2020.
2. Douglas R. Stinson and Maura B. Paterson, *Cryptography: Theory and Practice*, 4th Edition, Chapman & Hall CRC Press, 2019.
3. Nigel P. Smart, *Cryptography Made Simple*, Springer, 2016.
4. Ivan Niven, Herbert S. Zuckerman, and Hugh L. Montgomery, *An Introduction to the Theory of Numbers*, Wiley-India, 1991.
5. Israel N. Herstein, *Topics in Algebra*, 2nd Edition, Wiley India, 1991